

Complements of Operational Research

6.3 Queueing: finite population (M/M/R)

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When the “customers” are few

In 6.1 and 6.2 customers came from a huge population: one arrival did not change the arrival rate.

But in some systems the customers are **few and known**: the machines in a factory, the forklifts in a warehouse, the buses in a fleet. When one breaks down, it joins the technician’s “queue” — and can no longer break down.

State-dependent arrival rate

With K machines, n of which are broken down, the breakdown rate is

$$\lambda_n = (K - n) \lambda,$$

where λ is the breakdown rate of *one* working machine.

Notation: $M/M/R/GD/K/K$ — here we use R (not s) for the number of repairers, who are the servers; population (and capacity) K .

Typical question: how many technicians should be hired, if each idle machine costs money and so does each technician?

In class: the laundry with 5 machines and 3 mechanics (or one “super-mechanic”).

The example: forklifts in a warehouse

A warehouse has $K = 10$ forklifts. Each one breaks down, on average, once every 10 working days ($\lambda = 0.1$ per day); a repair takes, on average, 1.25 days ($\mu = 0.8$ per day); everything exponential.

Costs: each technician costs 150 €/day; each idle forklift costs 250 €/day (lost productivity, renting a replacement).

Question: 1, 2 or 3 technicians?

M/M/R with population K
($r = \lambda/\mu$)

$$P_n = \begin{cases} \binom{K}{n} r^n P_0, & n \leq R \\ \binom{K}{n} \frac{n!}{R! R^{n-R}} r^n P_0, & n > R \end{cases}$$

$$P_0 \text{ such that } \sum_n P_n = 1; \quad L = \sum n P_n;$$

$$L_q = \sum_{n>R} (n - R) P_n;$$

$$\lambda_{\text{ef}} = \lambda (K - L); \quad W = L / \lambda_{\text{ef}}.$$

Results

Technicians R	P_0	L (idle)	L_q	working	W (days)	cost (€/day)
1	0.122	2.97	2.09	7.03	4.23	$150 + 250 \times 2.97 = 893$
2	0.283	1.35	0.26	8.65	1.55	$300 + 250 \times 1.35 = \mathbf{636}$
3	0.305	1.14	0.04	8.86	1.29	$450 + 250 \times 1.14 = 736$

Decision

Two technicians: 636 €/day, 257 €/day less than with just one. With one technician, a broken-down forklift waits, on average, 3 days for repair ($W_q = W - 1/\mu \approx 2.98$).

The third technician still takes 0.2 forklifts out of idleness (50 €/day), but costs 150 €/day: it does not pay off.

Marginal rule, as in ch. 4: add capacity as long as the marginal saving is greater than the marginal cost.

Why not use the infinite-population model?

Temptation: “there are 10 forklifts at 0.1 breakdowns/day, so $\lambda = 1$ breakdown/day arrives; one technician does $\mu = 0.8$ ”. In an M/M/1, $\rho = 1.25 > 1$: **the system would be unstable** and the queue would grow without limit.

In reality, with one technician, there are on average 2.97 idle forklifts — bad, but stable. Broken-down machines do not break down again: the longer the queue, the fewer breakdowns arrive.

When to use which

- Population large compared with the number in the system (shop customers, calls): M/M/1, M/M/s.
- Small population (machines, vehicles, hospital inpatients): M/M/R with population K .

In the laundry from class, $\lambda_{\text{total}} = 5 \times 0.2 = 1$ per day against $\mu = 0.4$ for each mechanic: the error would be even larger.

```
from math import comb, factorial

def mmR(K, R, lam, mu):
    r = lam / mu
    w = [comb(K, n) * r**n if n <= R else
          comb(K, n) * factorial(n) * r**n / (factorial(R) * R**(n - R))
          for n in range(K + 1)]
    P = [x / sum(w) for x in w]
    L = sum(n * p for n, p in enumerate(P))
    Lq = sum((n - R) * p for n, p in enumerate(P) if n > R)
    return L, Lq, L / (lam * (K - L))      # idle, in queue, W

for R in (1, 2, 3):
    L, Lq, W = mmR(10, R, 0.1, 0.8)
    print(R, round(L, 2), round(150*R + 250*L))
# 1 2.97 893 | 2 1.35 636 | 3 1.14 736
```

For more realistic rules (technicians working shifts, repairs of almost fixed duration, priorities), the system is simulated — the code from 6.1 can be adapted by replacing the arrivals with the breakdowns of each machine.

Example 3 — the laundry

A laundry has 5 machines. Each machine breaks down, on average, once every 5 days and a mechanic repairs it, on average, in 2.5 days. There are currently 3 mechanics and it is possible to replace them with one “super-mechanic” who takes, on average, $5/6$ of a day to repair a machine. The super-mechanic’s wage equals that of the 3 mechanics. Breakdowns and repairs follow exponential distributions.

Should the laundry replace the 3 mechanics with the super-mechanic?

KEY TAKEAWAYS — 6.3

1. Finite population: $\lambda_n = (K - n)\lambda$; a machine that is broken down does not break down again.
2. M/M/R: P_n with $\binom{K}{n}r^n$ up to R and the factor $n!/(R!R^{n-R})$ above;
 $L_q = \sum_{n>R} (n - R)P_n$; $\lambda_{\text{ef}} = \lambda(K - L)$.
3. Decide the number of technicians with $\text{cost}(\text{technicians}) + \text{cost}(\text{idle machines})$ — marginal rule.
4. Do not use the infinite-population model when the number in the system is comparable to the population.

End of chapter 6. Next: 7 Inventory management.

References and resources

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Queueing Theory (finite calling population).
- Gross, D., Shortle, J. F., Thompson, J. M., Harris, C. M., *Fundamentals of Queueing Theory*, 4th ed., Wiley, 2008 — *machine repair problem*.
- Kasana, H. S., Kumar, K. D., *Introductory Operations Research*, Springer, 2004.

Resources

- Exercise sheet 6. `code/python/06_queueing/06_3_mmr/filas_mmr.py`.